

such as protocol analysers, network analysers, volt/amp/watt meters, logic analysers and signal generators are in large demand," adds Rajesh Suresh Joshi, manager-business development, Dynaloc (India).

There is tremendous potential for T&M in the defence industry too. "India's total spending on the defence industry is expected to reach \$36 billion by 2013. For supplies to defence, it is mandatory that a certain percentage of these be sourced locally even in the procurement deals signed with foreign suppliers," informs Neelam K. Kumar, executive director, Aplab Limited.

"The aerospace and defence sector would mostly be interested in high-end vector network analysers, spectrum analysers and signal generators up to 50 GHz," adds N.V. Valsalan, head-test & measurement and broadcast, Rohde & Schwarz India.

The increasing automation in various industries, including automotive, food and beverages, and textiles, will need sophisticated test instruments at various check points of every stage to minimise repairs afterwards in the production lines.

Devices available

Gautam Awasthi, general manager (marketing), electronic measurement group, Agilent Technologies India, classifies T&M devices into two broad categories based on their usage: Ad-

vanced/application-specific and basic.

Advanced or application-specific. Equipment such as network analysers, protocol testers, spectrum analysers, and cable and antenna testers cater to specific T&M requirements and therefore fall under application-specific category.

Cable and antenna testers measure cable losses, distance-to-fault functions, etc and are normally used in field installation and maintenance. Similarly, spectrum analysers are used in spectrum monitoring, checking the hopping and interfering signals in a particular band, and characterising any particular communication standard.

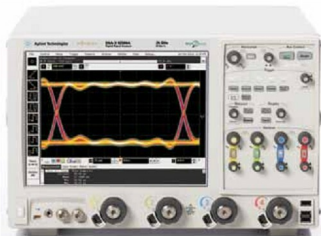
Basic. Equipment such as multimeters, power supplies, function generators, oscilloscopes and frequency counters come under basic category and can be seen on every electronics test bench. These instruments are normally used in general-purpose debugging of the device-under-test.

Recent trends driving innovations

Wireless. The combination of advanced technical capabilities brought about by digital RF and the increasing customer demand for more functionality and seamless mobility has led to a number of innovations in wireless communications. The latest advancements in T&M are mostly pertinent to WiFi test systems, SIM card tests and network testers.

"Latest technologies like LTE use MIMO antennae with 2x2 configuration enabling data rates of 100 Mbps in downlink and up to 50 Mbps in uplink. This means that today's test equipment should be able to simulate such scenarios and also handle multiple radio access technologies to facilitate testing," says Valsalan.

Embedded systems. Embedded



DSA-X 92504A digital signal analyser by Agilent Technologies

technology is being used to deliver new applications and services in a variety of industries including consumer electronics, industrial electronics, automotive, avionics, medical and communications. A major trend on the design side is energy efficiency. Integration of more switch-mode power supplies in embedded systems is bringing about test issues. The use of a switch-mode power supply obviously introduces switching into the end product.

Another key trend is dynamic power management. Embedded system designers using off-the-shelf processors will spend more time looking at power usage. Hence they need to learn how to better program the devices with programmable frequency and voltage scaling.

In general, embedded systems are rapidly becoming more complex and this applies not only to high-end handsets but also to low-cost children's toys. This, in turn, will lead to system-level test issues, which is eventually leading T&M vendors to package more tests into their instruments.

Serial data communication. The shift from parallel to high-speed serial data communications is one of the significant trends in recent times. So USB compliance testing is an important area for test and measurement. There are already some six billion USB devices worldwide. And with USB 3.0 poised to enter the design mainstream, it's a technology with huge growth potential.

Major applications

1. Communications (cellular test, wireless and wireline tests, optical)
2. Aerospace and defence (surveillance, signal intelligence)
3. Research & development (digital, RF, microwave, etc)
4. Security (surveillance, military communications)
5. Core electronics (data acquisition, control, automation, semiconductor, component test, nanotechnology, RF and microwave)

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